Dylan Dsouza

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Education

University of California San Diego (Provost Honors) San Diego, CA
Bachelor of Science in Data Science (GPA: 3.95) *Expected Mar 2027*

Technical Skills

Languages: Python, SQL, R, Java, C, C++, HTML, CSS, JavaScript
Libraries/Frameworks: NumPy, pandas, scikit-learn, PyTorch, TensorFlow, OpenCV, Plotly, LangChain, Jupyter, Git/GitHub
Core Competencies: Data Cleaning/Visualization, EDA, Web Scraping, A/B Testing, Financial Modeling, Time Series Analysis

Experience

Machine Learning Researcher Oct 2025 – Present
Engineers for Exploration *San Diego, CA*

- Engineering ensemble machine learning models (70% mIoU baseline) for automated mangrove mapping, developing end-to-end image segmentation pipelines to process satellite and drone imagery for pixel-level land-cover classification.
- Integrated 7 geospatial datasets (SpaceNet, xView, LandCover.ai) to optimize model architecture and feature engineering, accelerating ML-assisted labeling workflows across 4 environmental classes (road, building, water, mangrove).

AI Research & Development Intern Oct 2025 – Present
Computer Science & Engineering Society *San Diego, CA*

- Developing AgentSpace in a 9-person team, a platform leveraging hybrid GraphRAG architecture (Neo4j + graph traversal) and a game-theoretic personality framework to model professional decision-making, with initial focus on agentic tutoring systems.
- Designed an end-to-end RAG pipeline using LangChain and L2-normalized sentence embeddings (EmbeddingGemma-300M), implementing a hybrid retrieval algorithm that combines vector similarity with configurable graph-based context expansion.

Research Data Analyst May 2025 – Present
Scripps Institution of Oceanography *San Diego, CA*

- Engineered automated ETL pipeline processing 12,000+ georeferenced coral image annotations across 16 temporal snapshots (2014-2025), aggregating species-level observations into functional group metrics across 15 experimental reef sites.
- Performed time series analysis on 240 site-timepoint combinations, quantifying bleaching severity and recovery patterns for 9 coral genera under 5 experimental treatments, generating interactive visualizations that informed restoration strategies.

Quantitative Analyst Oct 2025 – Dec 2025
Sustainable Investment Group *San Diego, CA*

- Implemented automated Python data pipeline using FinBERT NLP, Yahoo Finance API, and Tavily to generate trading signals by quantifying lagged correlations between 25 weeks of news sentiment and 5+ market KPIs (returns, volatility, volume).
- Co-led quantitative investment thesis for Under Armour, integrating ARIMA time series forecasting, sentiment analysis, and DCF financial modeling to deliver data-driven sell recommendation in competitive 5-team stock pitch.

Market Research Intern Jul 2023 – Aug 2023
Bennett Coleman & Co. Ltd. (The Times of India) *Mumbai, India*

- Analyzed 56,000+ newspaper headlines using NLP pipelines (TF-IDF and BERT embeddings) and trained a multiclass classification model, achieving 85% accuracy across 5 market sectors (business, politics, technology, lifestyle, and social).
- Contributed to 3 internal meetings, assisting in evaluating data infrastructure and workflow optimization, including improvements to CRM, introduction of data lakes and data warehousing, and exploration of CDP and IP tracking systems.

Projects

NextMove: Predicting Next Workout Activity Nov 2025 – Dec 2025
Course Project — Recommender Systems *View Project*

- Engineered temporal, behavioral, and physiological features from 20,000+ sequential workout records in the Endomondo dataset, designing transition-based training instances to predict next activity type without look-ahead bias.
- Compared heuristic baselines (frequency, recency, weighted) against supervised ML models (Logistic Regression, Naïve Bayes, SVM, Random Forest), achieving 87% accuracy with Random Forest using chronological splits and balanced class weighting.

ENSOcast: Interactive Climate Analytics Dashboard May 2025 – Oct 2025
Personal Project (Presented at San Diego Undergraduate Tech Conference 2025) *View Project*

- Built a Streamlit dashboard with interactive Plotly visualizations for exploring 40+ years of ocean temperature data and El Niño-Southern Oscillation (ENSO) phases, enabling users to train and compare custom ML models.
- Developed and evaluated advanced ML models, including Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble Learning, using temporal, seasonal, and interaction-based features, achieving up to 82% accuracy in ENSO phase prediction.

Portfolio **Additional Projects** GitHub